ALGEBRA 1 · EXPONENTS AND SCIENTIFIC NOTATION

Exponent Rules Unpacker

Explore & Reflect

Name _____ Date _____ Period _____

Open **Exponent Rules Unpacker** on Mr. K's Classroom: Algebra 1 → Exponents and Scientific Notation → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Predict: $x^3 \cdot x^4 = x^{?}$

2. Predict the value of 2^0.

2 Explore use the tool and record what you see

Unpack the powers

Use the product, quotient and power rule tabs.

Expression	Written as factors	Number of x's left	Simplified
$x^2 \cdot x^3$			
$x^4 \cdot x$			
$\frac{x^5}{x^2}$			
$\frac{x^2}{x^5}$			
$(x^2)^3$			

Zero and negative exponents (base 2)

Each step down divides by 2.

Power	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}
Value	8					

3 Write the rule fill in each blank from what you observed

1. When you multiply powers with the same base, you _____ the exponents.

2. When you divide powers with the same base, you _____ the exponents.

3. When you raise a power to a power, you _____ the exponents.

4. Any nonzero number to the zero power equals _____.

5. x^{-n} = _____.

4 Check no tool: show your work

1. Simplify $a^6 \cdot a^2$.

2. Simplify $\frac{y^9}{y^4}$.

3. Simplify $(m^3)^4$.

4. Evaluate 5^{-2}.

5. Simplify $\frac{x^3}{x^3}$ two ways.

5 Reflect

Use the pattern table to explain why $3^0 = 1$.